

Gold Full Pipeline v1.4.2 Output Review

Scope: review of the latest full-chain run artifacts uploaded after the console returned qa_status = passed.

Verdict: technically integrated, but not yet commercially ready to freeze.

1. Executive decision

The pipeline now runs end-to-end and produces the expected major Gold artifacts. The scoring spine is coherent: the released score is 4.5 overall with criterion bands TR 4, CC 4, LR 5, and GRA 5; the verifier passed; the adjudicator confirmed the score; and the final score contract preserved the upstream score without modifying it.

However, the run is not ready for commercial freeze because several downstream learning outputs are still bridge-level or misaligned. The most serious issue is that the Priority Engine reports the primary limiter as UNKNOWN_SKILL. This breaks reliable routing into directive, feedback, practice, and learner-profile decisions.

2. What passed

Area	Status	Evidence
Submission	OK	Normalized WT2 essay with student_id and essay_id.
Detector/ErrorMap	OK technically	Detector failure_count = 0; ErrorMap error_count = 44.
Scoring	OK	Overall 4.5; TR 4, CC 4, LR 5, GRA 5.
Verifier	OK	pass / release.
Adjudicator	OK	confirmed; score_changed = False.
Score contract	OK	Integer criterion bands and half-band overall check passed.

3. Main blockers before freeze

Priority Engine mapping failure: Primary limiter is UNKNOWN_SKILL / UNKNOWN rubric, despite 34 evidence rows. This means the priority registry is not matching the standalone detector family codes or merged input format.

Directive has no real focus: Directive output has next_best_skill = null, next_best_capacity_domain = null, and recommended_service = null. This is a direct downstream symptom of the UNKNOWN_SKILL priority output.

Evaluator did not consume detector rows: Evaluator says detector_available = true but detector_row_count = 0. That means the detector-scorer assisted mode is not receiving or parsing detector evidence properly.

Evidence fusion is stale: The fusion artifact says LRET, Writing Coach, and Practice are not present even though they are generated later. The fusion stage is running too early or needs a late refresh.

Practice is not targeted: Practice output has primary_focus = null and produces mixed-review exercises. This is not acceptable for a Gold adaptive learning flow.

LRET missed FIX candidates: LRET produced 0 fix_units, although the essay clearly contains malformed lexical/semantic chunks. This suggests input conversion or LRET classification gating is too conservative in the full chain.

4. Educational-output quality

The Feedback Report is usable at a basic level: it identifies sentence_control, academic_style, and lexical_precision as top learning priorities. This aligns with the ErrorMap capacity counts: sentence_control 13, academic_style 12, lexical_precision 10, argument_development 8, cohesion_control 1.

The Writing Coach output is stronger than the bridge practice output because it selects a real microskill, arg_reason_generation, and a concrete move, Claim Plus Reason. This is pedagogically meaningful, but it is not yet fully aligned with the Priority Engine because the priority stage itself failed skill mapping.

The Learner Profile is structurally useful, but it creates a different next_best_action (argument_development -> writing_coach) while Skills Progress ranks sentence_control as the largest current gap. This may be explainable by service-routing logic, but the report should expose that logic clearly.

5. Freeze recommendation

Do not freeze the full Gold pipeline as a commercial release. Freeze only the orchestration milestone: v1.4.2 proves that an end-to-end run can execute and emit artifacts. The next production gate should be an output-quality gate, not another orchestration gate.

Recommended next version: v1.4.3 quality-gated integration. It should fix family-to-skill mapping, make directive focus non-null, pass detector rows into Evaluator, rebuild evidence fusion after downstream services, and ensure practice/LRET consume the same final focus/directive.

6. Go / no-go

Decision item	Status
Orchestrator pathing and subprocess execution	PASS
Score release contract	PASS
End-to-end artifact creation	PASS according to console output
Commercial learning-output quality	NO-GO
Safe to freeze full Gold version	NO
Safe to use for internal integration testing	YES

Note: The latest console output says qa_status = passed, but the latest QA report and manifest were not included among the uploaded artifacts for this review. This report therefore evaluates the uploaded stage artifacts and the console status together.